CHAPTER 2

Sketching in Code

In the last chapter, we managed to get our first sketch up and running with the famous Blink sketch, otherwise known as the "Hello world" example. This is often a good place to start because it has a long-standing tradition as the first program to write when learning a new language. In the case of programming hardware, we generally use the LED to blink hello. While that is a good beginning, we want to quickly get to some of the more interesting aspects of Arduino C, beginning with the focus of this chapter: the fundamental mechanics and control structures of sketching Arduino code. Our first project, RGB Blink, will explore the basic structure, syntax, and functions necessary for controlling the output of a three-color LED, taking our blinking LED to the next level.

We'll start with a primer on what sketching means and how we can put the philosophy of sketching in code to use. As we will do throughout this book, we will then jump right into our project example and once you have this project wired up and the sketch uploaded to your Arduino board, we'll back up and further explain how the sketch works and discuss most of the elements used in the project. In this way, when we look at each of the individual concepts in the chapter, you'll already know how they were put to use. In this chapter, we'll have a look at the basic anatomy of our project sketch and discuss how and why Arduino C does what it does. Don't worry if it doesn't make a tremendous amount of sense in the beginning, just work through it and it should get clearer as you go along.

What's needed for this chapter:

- Arduino Uno
- 5mm RGB LED with common cathode
- 1x 330 ohm and 2x 220 ohm ¼ watt resistors or similar
- Hookup wires
- Solderless breadboard

What is Sketching in Code?

The idea of sketching in code is a way of thinking about writing code as a simple intuitive process, just like drawing in a sketchbook. In this way, an Arduino program is called a sketch and is saved in a folder called a sketchbook. Sketching means we can get our hands dirty and quickly try out a new idea. It is a skill available to all of us, not just artists and designers, and neither is it limited to pens, paper, pencils, or napkins.